

THE INTERN DIARIES

Intern Report

Multitable Sales Analysis

WEEK 4 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

NORTHWIND (MULTI-TABLE ANALYSIS)

1. How many order-detail records and orders are present in the supplied Northwind data?

Answer: The analysis identified **26 order-detail records** in OrderDetails and **21 distinct orders** in Orders. The difference occurs because a single order can contain multiple product lines, so OrderDetails represents a lower-level transaction grain than Orders. The dashboard therefore uses distinct OrderID values when measuring orders rather than simply counting order-detail rows, which prevents one order with several products from being counted multiple times.

2. Are all order-detail records successfully matched with product information?

Answer: No. The relationship analysis found that only **9 of the 26 order-detail records** have a matching ProductID in the supplied Products table, giving a **product match rate of 34.6%**. The remaining **17 records** do not have corresponding product information, meaning their product price and therefore their sales amount cannot be reliably calculated from the supplied product master. The dashboard therefore presents product-based sales analysis as being based on the matched subset rather than treating the incomplete product relationship as complete.

3. Are the orders fully matched with the supplied customer information?

Answer: No. Only **3 of the 21 orders** have a CustomerID that is also present in the supplied Customers table, giving a **customer match rate of approximately 14.3%**. This indicates that the supplied customer master does not contain all of the customer IDs referenced by the Orders data. Rather than inventing missing customer information or dropping the unmatched orders, the analysis preserves the original transaction records and treats the missing customer relationships as a data-quality finding.

4. How was double counting prevented during the multi-table analysis?

Answer: Double counting was controlled by keeping OrderDetails as the lowest analytical grain and calculating sales at the product-line level as **Product Price × Quantity**. When measuring orders and customers, the analysis uses DISTINCTCOUNT logic so that an order containing several product lines remains one order and a customer appearing across multiple orders remains one customer. This prevents one-to-many joins from incorrectly multiplying order-level or customer-level totals and ensures that each metric is calculated at the appropriate level of detail.

5. What did the relational analysis reveal about the quality and usability of the supplied dataset?

Answer: The analysis showed that the seven Northwind CSV files are not fully aligned relationally: the source contains **26 order-detail records**, but only **9 have matching products**, while only **3 of 21 orders have matching customers**. This means the files can still be used to demonstrate multi-table SQL and Power BI analysis, but the resulting sales and customer findings must be clearly qualified as being based on the available matched records. The dashboard therefore combines business analysis with relationship-quality indicators so that users can distinguish between actual sales findings and limitations caused by incomplete source-table relationships.